

Food Delivery ETA Prediction

Using Machine Learning

Random Forest Regression Analysis

Report | 16 May 2026

1. Objective

The goal of this project is to build a machine learning model that accurately predicts food delivery time (in minutes) based on logistics-related features such as distance, traffic conditions, weather, and order characteristics. Accurate ETA prediction improves customer satisfaction and helps delivery platforms optimise their operations.

2. Dataset Overview

The dataset used is Food_Delivery_Times.csv, containing food order records with delivery metadata. Key features include:

- **Distance_km** — Distance between restaurant and delivery location
- **Traffic_Level** — Road traffic condition (Low / Medium / High)
- **Weather_Condition** — Weather at time of delivery
- **Time_of_Day** — Part of day (Morning / Afternoon / Evening / Night)
- **Vehicle_Type** — Mode of transport used by the delivery agent
- **Delivery_Time_min** — Target variable — actual delivery time in minutes

Data cleaning steps applied: removal of duplicate rows and rows with missing values, followed by label encoding of all categorical columns before model training.

3. Steps Performed

Step 1 — Data Upload & Loading

The CSV dataset was uploaded to Google Colab and loaded using pandas.

Step 2 — Library Setup

pandas, NumPy, Matplotlib, Seaborn, and scikit-learn were imported.

Step 3 — Data Cleaning

Duplicates were dropped and null values removed. Cleaned shape was verified with df.shape.

Step 4 — Exploratory Data Analysis (EDA)

Three visualisations were produced: a histogram of delivery time distribution, a scatter plot of Distance vs Delivery Time, and a box plot of Traffic Level impact on delivery time.

Step 5 — Feature Encoding

LabelEncoder was applied to all object-type columns to convert categorical values into numeric format.

Step 6 — Model Training

Features and target (Delivery_Time_min) were split 80/20. A RandomForestRegressor (random_state=42) was trained on the training set.

Step 7 — Model Evaluation

Predictions on the test set were evaluated using MAE, RMSE, and R² Score.

Step 8 — Export

The cleaned and encoded dataset was exported as cleaned_food_delivery.csv.

4. Model Performance Results

The Random Forest Regressor was evaluated on the 20% held-out test set:

Metric	Value	Interpretation
Mean Absolute Error (MAE)	7.05 minutes	On average, predictions are off by ~7 minutes
Root Mean Squared Error (RMSE)	9.81 minutes	Penalises larger errors; within acceptable range
R ² Score	0.764 (76.4%)	Model explains 76.4% of variance in delivery time

An R² of 0.764 indicates strong model performance for a real-world logistics dataset. The MAE of approximately 7 minutes is practically acceptable for a food delivery ETA system.

5. Key Findings from EDA

- Distance is the strongest predictor** — The scatter plot shows a clear positive correlation between Distance_km and Delivery_Time_min.
- Traffic conditions significantly impact ETA** — High traffic deliveries have noticeably higher median times and greater variance than Low or Medium traffic.
- Weather adds moderate uncertainty** — Adverse conditions such as rain and fog contribute to delays, though secondary to distance and traffic.
- Delivery time follows an approximately normal distribution** — The histogram shows a bell-shaped curve, indicating most deliveries complete within a predictable range.
- Random Forest captures non-linear relationships** — The model learned feature interactions (e.g., high traffic + long distance) that a linear model would miss.

6. Challenges in Real-World ETA Systems

Deploying ETA prediction in a live system introduces several complexities beyond static model training:

- Dynamic Traffic** — Real-time traffic changes minute-to-minute and cannot be fully captured by historical data alone.
- Weather Uncertainty** — Weather can shift rapidly during a delivery, requiring real-time weather feed integration.

- **GPS Inaccuracies** — Coordinate-based distance estimates may not reflect actual road distance or route complexity.
- **Route Diversions** — Road closures and accidents alter delivery times in ways not represented in training data.
- **Real-Time Processing** — Production systems must serve predictions in milliseconds for thousands of simultaneous orders.
- **Driver Behaviour** — Individual agent performance, route familiarity, and rest patterns add human variability.

7. Conclusion

This project successfully demonstrated that a Random Forest Regressor can predict food delivery ETA with meaningful accuracy, achieving an R^2 Score of 0.764 and a Mean Absolute Error of approximately 7 minutes. Distance and traffic level were the most influential features, with weather playing a secondary role.

The complete pipeline — from raw data ingestion through cleaning, EDA, feature encoding, model training, and evaluation — provides a solid foundation for a production-grade ETA prediction system. Future improvements could include real-time traffic and weather API integration, gradient boosting models such as XGBoost or LightGBM, and systematic hyperparameter tuning to further reduce prediction error.

References

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